Opie's technical work: the vertex cut

claude, 2026-11-07 · Opie's report of 2026-11-06 (order8_cut, SHA256SUMS checked), then his registry entries R9–R13.

In plain words first

- **Tim is the one hinge (PROVED).** The two worlds, the Chadverse core (7 posts) and Bronzina's coset (16 posts), share exactly one account, Tim, and Tim is on every post of both. Take Tim out and they fall apart; take anyone else out and nothing splits. That holds on Oct 16, on Nov 6, and on Tim's whole grid (1,757 and 1,752 posts): Tim is the only cut vertex every time.
- **The "8" is a count, not a group.** The 8 posts that left Tim's grid and the 8 that left LDA's grid are unrelated: no shared post, no overlapping dates. The only real order-8 group is $Z2^3$, in the core with B6 present, and it comes from which accounts the capture held, not from B6 itself.
- **One object, many pictures (§2).** The same 8-post object shows S5 × S6 symmetry as a matrix and only S3 as a concept lattice. Each picture hides something different.
- **What cut Bronzina off.** Her 6 posts leaving Tim's grid did not cut her from Tim on their own. Her name going off the other 10 did. The whole Tim–Bronzina link went, all 16 incidences together.
- **Superseded wording.** Opie's report says "removed / de-credited". Later the same day (registry R9 at the end), that was settled as one event: @sandrabronzina blocks us. Read §3 with R9.

Labels: PROVED = argued from the incidence facts · COMPUTED = his script's output · OBSERVED = in the captures · CONCLUSION-JOHN = John's call.

Order 8, the cut vertex, and what left Tim's grid

Opie (opencode), 2026-11-06. Offline; stdlib-only Python 3.14.4; no Sage, no numpy, no browser, no commits. Inputs: 20/20 files in `order8_input/` pass `sha256sum -c` (SHA256SUMS sha256 `1c70c2539f13…`). All six captures come from one viewpoint, including both 10-28 manifests. Every number below is reproduced by `code/order8.py`; witnesses are in `tables/` and `run.log`.

Answer first

1. **The "8" is a count, not a group order, and the two 8s are unrelated.** The 8 posts off Tim's grid are 6 Sandra Bronzina posts + 1 Laurent post + the B6 post (LDA-owned). The 8 posts off LDA's grid are 8 *borrowed* posts (7 owners; `king__mosiah` twice). The two 8-sets are disjoint. The only post common to TB's 8 and LDA's **12** removals is B6, and B6 sits in LDA's **4 own** removals, not in its 8 borrowed ones. The dates do not overlap (TB's 8: 2025-11-04 … 2026-07-16; LDA's 8: 2026-08-06 … 2026-10-27). **COMPUTED.**

1. **The 8-post incidence is highly symmetric, but not by an order-8 group.** 8 accounts × 8 posts; rooms `5+1+1+1`; identical columns `6+1+1`. |Aut_≤| = **120 = S5**; full |Aut| = **86,400 = S5 × S6**. The 2-part of 120 is 8; a Sylow 2-subgroup is **D4**, acting on 4 of the 5 B6-only accounts (fixing the fifth). That is a generic S5 fact, not a structure the number 8 selects. The reduced Galois lattice has **5 concepts** (the diamond M3), Aut **S3 (order 6)**, not 8. **COMPUTED.**

1. **The one genuine order-8 group in the core is Z2³, and it is a point-set fact, not a property of the block B6 as an edge.** On the LDA 10-16 core (B6 present) the rooms are `3,3,2`, |Aut_≤| = **72**, Sylow 2 = **Z2³** (order 8), acting by independent transpositions in `{LDA,LDC,TB}`, `{latterdaylogic,priesthoodmen}`, and `{mount_of_oliveslds,partisanrocktactical,roofkoreanactual}`. On the LDA 10-28 core (B6 absent, and the three B6-only accounts not in the captured point set) the rooms are `3,2`, |Aut_≤| = **12**, Sylow 2 = **C2 × C2**. If you delete B6 from the 10-16 point set but keep the three accounts, they become an isolated room of 3 and the group stays **72 / Z2³**. So the 12 → 72 jump is the capture's point set, not the block B6 as an incidence. The **abstract 14-point core** gives 36 both ways (Sylow C2×C2); it does not show the Z2³. **COMPUTED.**

1. **TB is the unique cut vertex of J, symbolically.** J = W1 ⊔ W2 with TB identified; W1 and W2 share only TB and no block. TB is on every block of both sides, hence adjacent to every other point on both sides. Removing TB separates the two non-empty sides; removing any other

point leaves TB adjacent to every remaining point, so the graph stays connected. **PROVED** from the incidence facts. It still holds on the 11-06 data (15 remaining J blocks; five isolated points H,I,J,K,SB, but TB is still the unique cut vertex). On the full TB grid (1,757 posts on 10-16; 1,752 on 11-06), Tarjan finds **TB the only cut vertex** on both. **PROVED / COMPUTED.**

1. **What removing the 8 posts did — and did not do.** On J the removal set is 7 blocks (B6 + the 6 SB posts); the Laurent post is not in J. B6 alone isolates the three B6-only accounts (mount, partisan, roof); the 6 SB posts alone do **not** cut SB–TB — SB is still on the other 10 S16 posts with TB in the 10-16 data (pair count 10). The actual 11-06 isolation of SB comes from a separate change: **SB is removed as a collaborator from those 10 surviving S16 posts** (10 de-crediting events). Laurent is removed the same way: 1 post removed (`Da3hv7oN5AZ`) plus 1 surviving post de-credited (`DZ3VA38h-la`). On the full TB grid, removing the 8 posts alone removes mount, partisan and roof but leaves SB and Laurent present; the 11-06 grid has all five absent. **COMPUTED.**

1. **The four B6 accounts other than LDA and TB.** LDC stays on the core grid (12 posts with TB on 11-06); mount, partisan and roof lose their only post. So three of the four are separated, LDC is not. **COMPUTED.**

1. Order 8: every place an 8 (or an order-8 group) arises

1.1 The two 8s

side	8 posts	composition	dates
TB grid, 10-16 → 11-06	1,757 → 1,752	6 sandrabronzina + 1 laurentheinsider + 1 B6 (latterdayavenger-owned)	2025-11-04 ... 2026-07-16
LDA grid, 10-28 → 11-06	8 borrowed (of 12 left; 4 LDA-owned); grid 333 → 332	king__mosiah ×2; elders_in_the_endzone, mormon.marxist, date.celestial, ben.potzsch, apotheosis_memes, befaithfulpodcast ×1	2026-08-06 ... 2026-10-27

The two 8-sets share **no post**. The only post common to TB's 8 and LDA's **12** removals is **B6** (`DW99EnrgVYo`), and it is one of LDA's **4 own** removals, not one of its 8 borrowed. The two 8s are therefore different events with different owners, different dates and different mechanisms. **COMPUTED.**

A uniform-post null makes the "all 6 Sandra posts in the 8" event $p \approx 6.9 \times 10^{-16}$, but the assumption is wrong: removal is account-level (6 of 6 Sandra posts; the Laurent post and the B6 post), not 8 independent draws. We do not use that null as evidence; it assumes away the only structure present.

1.2 The structures and their groups

`Aut_≤` = automorphisms preserving the block order (each block fixed), equal to the product of |room|! over rooms of points with identical rows. `Aut` = the full incidence automorphism group (points and blocks may permute). The Sylow 2 column is the Sylow 2-subgroup of `Aut_≤`.

structure	v	b	rooms	`Aut_≤`	`Aut`	Sylow 2 of `Aut_≤`	concepts
TB8 (8 accounts × 8 posts)	8	8	5,1,1,1	**120 = S5**	**86,400 = S5×S6**	**D4 (order 8)**	5
LDA8 (15 accounts × 8 borrowed posts)	15	8	4,3,2,1…	**288**	**1,152**	order 32 (D4×C2×C2)	10
core B1–B7 (abstract, 14 points)	14	7	3,3,1…	**36**	**36**	C2×C2	16
core minus B6 (abstract)	14	6	3,3,1…	**36**	**36**	C2×C2	15
core B1–B5 (abstract)	14	5	4,3,2,1…	288	288	order 32	12
core B1–B5 (LDA 10-28 capture)	10	5	3,2,1…	**12**	**12**	**C2×C2**	12
core B1–B6 (LDA 10-16 capture)	13	6	3,3,2,1…	**72**	**72**	**$Z2^3$ (order 8)**	13
core B1–B6 (LDA 10-16) minus B6, same 13 points	13	5	3,3,2,1…	**72**	**72**	**$Z2^3$ (order 8)**	12
J 10-16	28	23	3,3,2,2,1…	**144**	≥ 103,680 (6! block class; not enumerated)	order 16 ($C2^4$)	32
J after B6+6SB (10-16 credits)	28	16	3,3,2,2,1…	**144**	**1,728**	order 16 ($C2^4$)	31
J actual 11-06 (15 blocks)	28	15	5,3,2,2,2,1…	**5,760**	**69,120**	order 128 ($D4×C2^4$)	27

Reading the table:

- **TB8 gives D4**, not Z8/Z4×Z2/$Z2^3$/Q8. D4 is the Sylow 2-subgroup of S5; it permutes 4 of the 5 identical B6 co-posters and fixes the fifth. There are 5 choices of fixed point and several Sylow subgroups; no canonical one.
- **LDA8 gives order 32**, so it contains D4 subgroups but its Sylow 2 is not order 8. Its block-8 room is S4 (D4 inside), its other rooms S3 and S2.
- **Core B1–B6 (LDA capture) is the only exact order-8 Sylow of a non-D4 type: $Z2^3$.** It is the 2-part of S3 × S2 × S3, i.e. one transposition in each of the three rooms. The subgroup is not unique (3 choices of pair in each S3), but the type $Z2^3$ is determined. The 12 → 72 jump is the 10-28 → 10-16 capture point set: the 10-28 capture does

not contain the three B6-only accounts (they are on B6 only). Deleting B6 from the 10-16 point set but keeping them leaves 72 (they become an isolated room of 3); deleting them as well gives

- So the $Z2^3$ is a point-set/representation fact, not a property of B6 as an edge.
- **The abstract core does not show this**: it already has an H,I,J room on B6, so removing B6 leaves a room of isolated points and the product stays 36. The $Z2^3$ is an artefact of the LDA-capture account set, not of the abstract core lattice.
- **J** has 2-part 16 or 128, never 8; the ordered group is 144 on J and 5,760 on the 11-06 J (the removals create five isolated points, a 5! factor).
- **No Z8, Z4×Z2 or Q8 appears** as the Sylow 2-subgroup of any ordered group here. The types are $C2^k$, D4, and products such as $D4\times C2^k$.
- **No 8-element lattice**: the TB8 Galois lattice has 5 concepts (M3; Aut S3), the core B1–B7 16, core minus B6 15, core B1–B5 12, core B1–B6 13, core B1–B6 minus B6 12, LDA8 10, J 32. The number 8 is not a lattice order either.

Conclusion of §1 (COMPUTED): the two 8s are unrelated counts. The 8-post object's symmetry is S5 × S6, not an order-8 group. The only exact order-8 Sylow with a different type is $Z2^3$ in the core **with B6** in the LDA 10-16 capture (the three B6-only accounts are in the point set); deleting B6 but keeping them leaves the same group. If John is looking for an order-8 group tied to B6, that is the one; if he is looking for an 8 tied to the 8 removed posts, there is none.

2. Differing representations — same object, the TB8 incidence

Object: 8 accounts × 8 posts, posts in date order. Incidences: TB 8, SB 6, Laurent 1, the five B6 accounts 1 each (20 incidences).

representation	what it shows	what it hides
incidence matrix (`tables/tb8_incidence.csv`)	TB row all 1s; SB row six 1s; Laurent one; five B6 rows one; six identical SB columns; B6 column weight 6	owner/date unless labelled; the six identical columns are six distinct dates; the block multiplicity (the matrix gives six columns, the ordered group treats them as ordered); the clique expansion of B6
Levi (bipartite) graph (`tables/tb8_levi_edges.tsv`)	8 account vertices + 8 post vertices, 20 incidences; the six SB post vertices are twins; the five B6 account vertices are twins; TB is the hub	which of the six SB posts is which (if unlabelled); the date order; the difference between the five B6 accounts

account graph / 2-section (`tables/tb8_account_edges.tsv`)	TB adjacent to all 7; SB and Laurent pendant to TB; the B6 accounts form a K6 with TB; 17 edges	the sixfold multiplicity of SB–TB (one edge, not six posts); the time order; the sixfold block symmetry S6
hypergraph (blocks as sets)	six copies of `{SB,TB}`, one `{LDA,LDC,mount,partisan,roof,TB}`, one `{LI,TB}`	which copy is which; the dates; the fact that the copies are distinct posts (the set of sets collapses duplicates)
permutation representation of Aut on points	image **S5** (order 120); the five B6 accounts are one orbit; TB, SB, Laurent fixed	the block action; the kernel S6
permutation representation of Aut on blocks	image **S6** (order 720); the six SB posts are one orbit; B6 and the Laurent post fixed	the point action; the kernel S5
ordered (Holland) representation `Aut_≤`	**S5** (120), blocks fixed in date order; the only symmetry is the five B6 accounts	the block permutations (the six SB posts are distinct dates and cannot be permuted)
concept/Galois lattice (`tables/tb8_concepts.tsv`)	**5 concepts** (M3: top, three atoms SB-type / B6-type / Laurent-type, bottom); Aut **S3 (6)**	the multiplicities (five B6 accounts, six SB posts) and the time order; it collapses duplicates, so it *loses* S5×S6
dual/transpose	the six SB posts become a room of 6; the five B6 accounts a room of 5	nothing new, but makes the S6 a point room and the S5 a block room

The lesson: the incidence matrix and Levi graph show the multiplicity; the account graph and hypergraph collapse it; the permutation representations name the symmetry; the ordered representation removes the block permutations; the concept lattice removes the multiplicities and keeps only the closure structure. That is why the same data can look like "S5 × S6" in one representation and "5 concepts, S3" in another.

3. The cut vertex, symbolically

Definitions. S = (P,B,I), x incident with b when x is owner or listed collaborator. G(S) is the account graph: V=P, edge {x,y} when some block contains both. A cut vertex is v with G−v having more components than G.

PROVED (unique cut vertex of J). In J = W1 ⊔ W2 with TB identified: W1 and W2 share exactly the point TB and no block (the 10-31 J: core 7 posts, S16 16 posts, 28 accounts, 93 edges). TB is on all 7 W1 blocks and all 16 W2 blocks, so TB is adjacent to every other point on both sides. Removing TB separates the two non-empty sides. For any x ≠ TB, x lies on one side and is adjacent to TB there; after removing x, TB remains and is adjacent to every remaining point on both sides, so G−x is connected. Hence TB is the unique cut vertex. The computed check agrees: 28 accounts, 23 posts, 93 edges, G−TB has two components of sizes 14 and 13, cut vertices `['timballard89']`, no bridges.

PROVED on the 11-06 data (J_1106). The 15 J blocks still on the 11-06 TB grid are core B1–B5 + the 10 surviving S16 posts. The core points are the anonymised A–K/LDC/TB/LDA of the 10-31 lattice; the 11-06 capture has no key, so the core side is held fixed for B1–B5 (TB's presence on those blocks is checked in B), while the S16 side uses the 11-06 credits. TB is on all 15; W1 and W2 still share only TB. The same cone proof applies. The graph now has five isolated points (W1:H, W1:I, W1:J, W1:K, sandrabronzina), so it is disconnected; removing TB still raises the component count (6 → 12), and no other point is a cut vertex.
COMPUTED: 28 points, 15 blocks, 61 edges, 6 components `23,1,1,1,1,1`; G−TB 12 components `9,5,3,2,1,1,1,1,1,1,1,1`; bridges `ivetteavaria–TB`, `larinoar–TB`, `maaydelara–TB`.

COMPUTED on the full grid. On the 1,757-post 10-16 grid: 327 accounts, 822 edges, TB degree 326, cut vertices `['timballard89']`, 58 bridges. On the 1,752-post 11-06 grid: 323 accounts, 786 edges, TB degree 322, cut vertices `['timballard89']`, 63 bridges. TB is the unique cut vertex on both.

3.1 What the 8 posts did to J

The 8 posts are 6 SB + 1 Laurent + B6. Only 7 of them are blocks of J: B6 and the 6 SB posts. The Laurent post is outside J.

object	v	b	edges	components	G−TB	cut vertices
J 10-16	28	23	93	1 (28)	14,13	TB
J after B6+6SB (10-16 credits)	28	16	81	4 (25,1,1,1)	14,10,1,1,1	TB
J on 15 blocks (B7 not on TB grid), 10-16 credits	28	15	75	5 (24,1,1,1,1)	14,9,1,1,1,1	TB
J actual 11-06 (B credits)	28	15	61	6 (23,1,1,1,1,1)	9,5,3,2,1×8	TB

- **Block-only effect:** removing B6 isolates the three B6-only accounts (W1:H, W1:I, W1:J = mount, partisan, roof). The 6 SB posts do **not** isolate SB: in the 10-16 credits SB is still on the other 10 S16 posts with TB (SB–TB count 10). So the 7-block set is a cut set (it isolates H,I,J) but not a minimal one; B6 alone does the isolating, and the 6 SB blocks alone do not.
- **Actual 11-06 effect:** SB is removed as a collaborator from those 10 surviving S16 posts (10 de-crediting events) and Laurent from its surviving post `DZ3VA38h-la` (1 de-crediting event), on top of the 6 SB + 1 Laurent post removals. That is what isolates SB and Laurent. The ordered account graph then has five isolated points (H,I,J,K,SB) and three bridges to TB.
- **Pair effect** (`tables/removal_effects.tsv`): of the 17 pairs carried by the 8 posts, only LDA–TB, LDC–TB and LDA–LDC survive in J after the block removal (counts 6, 6, 6; 5 each in the actual 11-06 J; 11, 12, 7 on the

full 11-06 grid). SB–TB is 10 in J after the block removal, 0 in the actual 11-06 J and 0 on the full 11-06 grid — the drop is the de-crediting, not the 6 post removals. All pairs involving mount, partisan or roof and all Laurent–TB are 0 after removal.

3.2 Edge cut / vertex cut, said exactly

- The 8 removed **posts** are not vertices of the account graph, so they are not a vertex cut of it. In the Levi graph, B6 is a cut vertex (it isolates H,I,J); the 6 SB block-vertices are not a cut; the Laurent block is outside J. The 8-set is not minimal and is not one structural cut.
- On the account graph, the 8-post removal alone is **not** an edge cut that separates SB from TB (SB still has 10 S16 posts with TB in the 10-16 credits). The actual 11-06 change set — 6 SB posts + 10 SB de-crediting + 1 Laurent post + 1 Laurent de-crediting — is an edge cut that isolates SB and Laurent. SB is **not** a leaf in J: she shares the 16 S16 blocks with the other guests, so the cut is the set of her 16 block incidences (6 removed posts + 10 de-credited), not a single SB–TB edge. On the full grid the same holds (SB degree 14 on 10-16).
- The single account **TB** is a vertex cut of size 1 of J and of the full grid (computed). That is the cut that matters; the 8 posts are not a second one.

4. What cannot be told

Instagram does not record who removes a post, who removes a collaborator, or who archives. All the captures are from the same viewpoint, which reduces but does not eliminate visibility differences. "Left TB's grid" is a set difference between two snapshots; it is consistent with owner deletion, archiving, a platform takedown, or a change in what that viewpoint can see. B6 being off TB's 11-06, LDA's 10-28 and 11-06, and LDC's 10-28 captures (and claude reports it off LDA's 10-17) is consistent with any of those; it does not identify an actor. The statement that LDA's 8 borrowed posts are still on their owners' grids with LDA listed is claude's check, not re-derivable from the files held here. The SB and Laurent de-crediting events are visible as changed collaborator lists; whether the owner removed them, they left, or the platform did it cannot be told from these captures.

5. Reproduce

```
python3 -I code/order8.py        # all numbers -> run.log + tables/
python3 code/make_sums.py        # INPUTS.sha256 + SHA256SUMS
```

`tables/`: tb8_posts, tb8_incidence, tb8_levi_edges, tb8_account_edges, tb8_concepts, lda8_incidence, core_variants, j_variants, j_credit_changes,

j_concepts, removal_effects, order8_search, grid_accounts_8, grid_account_changes_8.

No commits; inputs untouched. In-house until John says otherwise.

— Opie (opencode)

Registry entries R9–R13 (Opie, 2026-11-06b)

R9 — the SB block (CONCLUSION-JOHN + OBSERVED). @sandrabronzina is not visible to us. John's call: the account is active and blocks us. So SB's 16 lost links are one account-level event — a block — not 6 removals plus 10 de-creditings. Replace any wording that says she removed posts or that she was removed from them.

- **OBSERVED:** on 2026-10-16 she is on all 16 S16 posts; by 2026-11-06 she is on none; her 6 own posts are off Tim's grid; her name is off the 10 surviving posts.
- **CONCLUSION-JOHN:** the cause is a block by an active account.
- **Not computed:** from our side a block, an untag, a collaborator removal or a mix all fit the same metadata. John's call identifies the cause; the math does not.

R10 — block bracket (OBSERVED, corrected). Collaborator-visible on Oct 16; on Tim's Oct 28 capture her May–June own posts are absent and she is a collaborator on none; her @-credits on B03, B09 and B15 remain only because they are caption text, which a block does not hide. So the block falls **between Oct 16 and Oct 28**. My earlier report said "visible Oct 28": that is true only of the lingering caption credits, not of her presence on the posts; correct it.

R11 — laurentheinsider (OBSERVED). Visible from one of our viewpoints, hidden from the other since Oct 16; its post `Da3hv7oN5AZ` is gone from its own grid too; its de-credit on `DZ3VA38h-la` is a viewer effect, not a finding. (The post's absence is read on the account's own grid; the missing credit is read from the viewpoint that cannot see the account.)

R12 — J on Nov 6 (COMPUTED, claude's check, published). From the published matrices + `s16_status.csv`: 29 accounts, 15 posts, 65 lines, 6 pieces (one big piece; H, I, J, K, sandrabronzina with no line); Tim is still the only cut vertex. Opie's 61 lines used **28 points, without chemalibreria**; chemalibreria adds 4 lines (R8 above), giving 65. Witness: `docs/data/bronzina-cut/bronzina_cut_check_output.txt`.

R13 — casting (CONCLUSION-JOHN). In the one-monkey frame (M5: zero monkeys), timballard89 and sandrabronzina are **co-bots**, seated by expofamily.cl, the arranger (M4). The structure under the casting is computed (R7/R8); the casting is John's. Do not say, as in M5:

- "the math proves there is no human";
- "the math proves expofamily.cl is a robot";
- "the math proves the others are software";
- "the cut vertex is the operator" (TB is the most central account, not the same property as the operator/arranger).

Also do not say "the math identifies who blocked whom": the block is observed and named by John (R9), not computed.

— Opie